

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following:		12
A)	Given point P (-2,6,3), and vector $\mathbf{A} = y \mathbf{a}_x + (x+z) \mathbf{a}_y$, Express P and A in Cylindrical and Spherical Coordinates.	Understand	6
B)	Three field quantities are given as $\mathbf{P} = 2\mathbf{a}_x - \mathbf{a}_z$, $\mathbf{Q} = 2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z$, $\mathbf{R} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$. Determine (i) $(\mathbf{P} + \mathbf{Q}) + (\mathbf{P} - \mathbf{Q})$ (ii) $(\mathbf{Q} \bullet \mathbf{R} \times \mathbf{P})$ (iii) $(\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}))$	Understand	6
C)	Explain the term Curl of a vector and hence derive the Stokes Theorem	Understand	6
D)	Explain the term Divergence of a vector field and hence derive the Divergence Theorem. Determine the Divergence of the vector field $\mathbf{A} = YZ \mathbf{a}_x + 4XY \mathbf{a}_y + Y \mathbf{a}_z$ at a given point (1,-2,3)	Understand	6
Q.2	Solve Any Two of the following.		12
A)	Derive an Expression for Electric Field Intensity $\mathbf{E}$ due to Infinitely Long Straight Uniformly Charged Wire	Understand	6
B)	State and explain : Ampere's Circuit law and Maxwell's Continuity equation	Remember	6
C)	Find total charge enclosed by the volume with charge density $\rho_v = 4xyz^2$, For $0 \leq \rho \leq 2$, $0 \leq \theta \leq \pi/2$ and $0 \leq z \leq 3$.	Understand	6
D)	State and explain the Coulomb's law and the superposition principle applied for the static charge distribution.	Remember	6
Q. 3	Solve Any Two of the following.		12
A)	State and Explain the Boundary conditions for medium separating : Dielectric- Dielectric and Conductor – Dielectric	Remember	6
B)	State and explain the gradient of a scalar field. Show that $\mathbf{E} = - (\text{Grad } V)$	Understand	6

where V is a scalar function and hence determine the gradient of the scalar field $U = x^2 y + x y z$

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| C) | State and explain the basic Maxwell's Equations in its point and integral form with their proper significance. | Remember | 6 |
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Q.4	Solve Any Two of the following.	12
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| A) | A transmission line cable has following primary constants. $R = 75 \Omega/\text{km}$, $G = 60 \mu\text{mho}/\text{km}$, $L = 1.7 \text{ mH}/\text{km}$ and $C = 0.075 \mu\text{F}/\text{km}$, at a signal of 2kHz, calculate (i) characteristics impedance, (ii) attenuation constant (iii) phase constant (iv) wavelength (v) velocity and (vi) time taken by signal to travel 100 km along line. | Understand | 6 |
| B) | Derive the expression for Electric Potential Energy and Electrical Potential. | Remember | 6 |
| C) | Define the following terms with reference to EM Wave:
(i) Propagation constant (ii) characteristic impedance and (iii) VSWR | Remember | 6 |
| D) | State ohms law in EM field and prove the equation for ohms law
$J = \sigma E$, where J is conduction current density in amp per meter square ,
σ (Sigma) is the conductivity and E is electric field strength in volt per meter. | Understand | 6 |

Q. 5	Solve Any Two of the following.	12
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| A) | A uniform plane wave of 50 Hz propagating through Copper conductor having $\sigma = 6.8 \times 10^7 \text{ mho}/\text{m}$, $\epsilon_r = 1$, $\mu_r = 1$. Find (i) attenuation constant, (ii) phase constant, (iii) propagation constant, (iv) intrinsic impedance, (v) wavelength and (vi) phase velocity. | Understand | 6 |
| B) | Given the circular plate parallel capacitor with radius of 5 cm where the distance of separation between the plates is 2cm and connected to a battery of 200 Volt. Find the value of capacitor and the energy stored in the capacitor. | Understand | 6 |
| C) | Show that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within volume v minus the ohmic losses. | Understand | 6 |
| D) | Derive the equation for the uniform EM plane waves for the free space. | Remember | 6 |

***** End *****